

# Change of Numeraire

Let  $S_T$  be an asset in units of cash at time  $t$ .

The value of cash changes over time.

A common way to approach the modeling is to represent the price in terms of the value of the bank account,  $B_t$ .

We start with  $B_0 = 1$ , and suppose it is invested at the  $r_f$ ,

then  $B_t = B_0 e^{r_f t}$  and  $B_T = B_t e^{r_f(T-t)}$   $t < T$ ,

if the  $r_f$  is not constant, then  $B_T = B_t e^{\int_t^T r_s ds}$ .

We can scale the value of the asset by the value of the bank account by  $\frac{S_t}{B_t}$ . For  $n$  assets let it be  $\frac{S_t^1}{B_t}, \dots, \frac{S_t^n}{B_t}$ .

If we have an option with value  $V_t$ , we can scale it.

And then we know that from the risk neutral valuation, the ratio is a martingale under the risk neutral measure

$$\frac{V_t}{B_t} = E^Q \left[ \frac{V_T}{B_T} \mid \mathcal{F}_t \right], \text{ we will denote it by } E^Q. \quad \leftarrow \text{we have a lot of measures}$$

Similar to the risk neutral valuation, the ratio of the option scaled by the value of the numeraire asset will be a martingale under a different probability

measure

$$\frac{V_t}{N_t} = E^N \left[ \frac{V_T}{N_T} \mid \mathcal{F}_t \right].$$

To know the value of  $V_t$ , we use

$$E^Q \left[ X \mid \mathcal{F}_t \right] = \frac{E^N \left[ X \frac{dQ^B}{dQ^N} \mid \mathcal{F}_t \right]}{E^N \left[ \frac{dQ^B}{dQ^N} \mid \mathcal{F}_t \right]}$$

which is similar to the  $P(A|B)$ , then:

$$V_t = B_t E^B \left[ \frac{V_T}{B_T} \mid \mathcal{F}_t \right],$$

$$V_t = N_t E^N \left[ \frac{V_T}{N_T} \mid \mathcal{F}_t \right]$$

$$V_t = B_t \frac{E^N \left[ \frac{V_T}{B_T} \frac{dQ^B}{dQ^N} \mid \mathcal{F}_t \right]}{E^N \left[ \frac{dQ^B}{dQ^N} \mid \mathcal{F}_t \right]} = N_t E^N \left[ \frac{V_T}{N_T} \mid \mathcal{F}_t \right].$$

$$V_0 = B_0 \frac{E^N \left[ \frac{V_T}{B_T} \frac{dQ^B}{dQ^N} \mid \mathcal{F}_0 \right]}{E^N \left[ \frac{dQ^B}{dQ^N} \mid \mathcal{F}_0 \right]} = N_0 E^N \left[ \frac{V_T}{N_T} \mid \mathcal{F}_0 \right].$$

The expectations now are under the same probability measure. If the two expected values are equal we can deduce that the expression will also be equal

$$B_0 \frac{\cancel{V_T}}{B_T} \frac{dQ^B}{dQ^N} = N_0 \frac{\cancel{V_T}}{N_T}$$

$$\frac{dQ^B}{dQ^N} = \frac{B_T / B_0}{N_T / N_0}$$

It links the ratios of the probabilities, to the ratios of asset prices.

# Margrabe model (option formula)

Change of numeraire. Trick to change the measure.

$$C_T = \max\{S_2(\tau) - S_1(\tau), 0\}$$

Payoff. allows you to exchange  $S_1$  and  $S_2$ .  
Pays no dividends

Pricing equation. Risk neutral pricing formula

The value of the option with respect to some numeraire which will be on the denominator must be equal to the risk neutral expectation given by the numeraire of the payoff / numeraire at maturity.

$S_1$  ... numeraire

By risk neutral valuation formula

$\frac{C_0}{S_1(0)}$  = risk neutral expectation bounded to the asset of the terminal period divided by  $S_1(\tau)$

$$\frac{C_0}{S_1(0)} = \tilde{\mathbb{E}} \left[ \frac{\max\{S_2(\tau) - S_1(\tau), 0\}}{S_1(\tau)} \right]$$

$$C_0 = S_1(0) \tilde{\mathbb{E}} \left[ \frac{\max\{S_2(\tau) - S_1(\tau), 0\}}{S_1(\tau)} \right]$$

$$\text{let } Z_t = \frac{S_2(\tau)}{S_1(\tau)}$$

$$C_0 = S_1(0) \tilde{\mathbb{E}} \left[ \max\{Z_\tau - 1, 0\} \right]$$

expectation under the risk neutral measure

we have no idea of  $Z_t$ , hahaha!

Motions of  $S_1$  and  $S_2$ . They come from a differential equation?

$$dS_i(t) = \underbrace{\mu_i S_i(t) dt}_{\text{deterministic}} + \underbrace{\sigma_i S_i(t) dW_i(t)}_{\substack{\text{differential of the} \\ \text{brownian motion}}}$$

Lets say they are correlated.

Lets focus on the dynamics of  $T$  on the "empirical measure" and transform it to the risk neutral measure.

Ito's lemma:

$$\begin{aligned} \frac{dZ_t}{Z_t} &= \frac{dS_2(t)}{S_2(t)} - \frac{dS_1(t)}{S_1(t)} - \frac{dS_2(t) dS_1(t)}{S_2(t) S_1(t)} - \left( \frac{dS_1(t)}{S_1(t)} \right)^2 \\ &= \underbrace{\mu_2 dt + \sigma_2 dW_2(t)}_{\text{Diffusion terms}} - \underbrace{\mu_1 dt + \sigma_1 dW_1(t)}_{\text{Diffusion terms}} - \underbrace{(\mu_2 dt + \sigma_2 dW_2(t))(\mu_1 dt + \sigma_1 dW_1(t))}_{\text{Diffusion terms}} - \sigma_1^2 dt \\ &= (\mu_2 - \mu_1 - \sigma_1^2) dt + \underbrace{\rho \sigma_1 \sigma_2 dt + \sigma_2 dW_2(t) - \sigma_1 dW_1(t)}_{\text{Diffusion terms}} \end{aligned}$$

$$= (\mu_2 - \mu_1 - \sigma_1^2 + \rho \sigma_1 \sigma_2) dt + \underbrace{\sigma_2 dW_2(t) - \sigma_1 dW_1(t)}_{\text{Diffusion terms}}$$

we want to get rid of this B.M.

$$W_i \sim \mathcal{N}(0, \sigma_i^2 dt)$$

The diff is also normal

$$\{W_2 - W_1\} \sim \mathcal{N}(0, (\sigma_1^2 + \sigma_2^2 - \rho \sigma_1 \sigma_2) dt)$$

lets denote by  $\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - \rho \sigma_1 \sigma_2}$  to create an artificial brownian motion in terms of  $\sigma$

$$= (\mu_2 - \mu_1 - \sigma_1^2 + \rho \sigma_1 \sigma_2) dt + \sigma dW(t)$$

under risk neutral measure this term is a martingale

lets invoke The Girsanov theorem for brownian motion  
(change of probability measure)

$$= (\mu_2 - \mu_1 - \sigma_1^2 + \rho \sigma_1 \sigma_2) dt + \sigma dW(t)$$

under risk neutral measure this term is a martingale

lets invoke The Girsanov theorem for brownian motion  
(change of probability measure)

$$I_t = \exp \left\{ \int_0^t -\frac{1}{\sigma} (\mu_2 - \mu_1 - \sigma_1^2 + \rho \sigma_1 \sigma_2) dW(s) - \frac{1}{2} \int_0^t \frac{1}{\sigma^2} (\mu_2 - \mu_1 - \sigma_1^2 + \rho \sigma_1 \sigma_2)^2 ds \right\}$$

That's the redundant "redundant" derivative

The transformation kernel:

$$\phi_t = -\frac{1}{\sigma} (\mu_2 - \mu_1 - \sigma_1^2 + \rho \sigma_1 \sigma_2)$$

Change of measure theorem, the brownian motion under the empirical measure is:

$$dW(t) = \phi_t dt + d\tilde{W}(t)$$

↑  
new brownian motion under risk neutral measure.

With this we can kill the drift  $\tilde{W}$  (in  $\frac{dZ_t}{Z_t}$ )

$$dW(t) = \phi_t dt + d\tilde{W}(t)$$

$$dW(t) = \left( -\frac{1}{\sigma} (\mu_2 - \mu_1 - \sigma_1^2 + \rho \sigma_1 \sigma_2) \right) dt + d\tilde{W}(t)$$

$$\begin{aligned}\frac{dZ_t}{Z_t} &= (\mu_2 - \mu_1 - \sigma_1^2 + \rho\sigma_1\sigma_2)dt + \sigma dW(t) \\ &= (\mu_2 - \mu_1 - \sigma_1^2 + \rho\sigma_1\sigma_2)dt + \sigma \left[ \left( -\frac{1}{\sigma} (\mu_2 - \mu_1 - \sigma_1^2 + \rho\sigma_1\sigma_2) \right) dt + d\tilde{W}(t) \right] \\ &= \sigma d\tilde{W}(t)\end{aligned}$$

↑  
Wiener differential under risk neutral measure

$I_t$  is the behavior of the ratio of two stocks under risk neutral measure.

We want to know the distribution of that

$$\frac{dZ_t}{Z_t} = \sigma d\tilde{W}(t)$$

$$dZ_t = Z_t \sigma d\tilde{W}(t)$$

sup. It is an exponential solution, because it is similar to stock motion in Black-Scholes.

$$\text{let } Y = \ln(Z_t)$$

$t$  is constant, so this is 0

$$dY = \frac{\partial Y}{\partial Z_t} dZ_t + \frac{\partial Y}{\partial t} dt + \frac{1}{2} \frac{\partial^2 Y}{\partial^2 Z_t} (dZ_t)^2$$

$$dY = \sigma d\tilde{W}(t) - \frac{1}{2} \frac{1}{Z_t^2} (dZ_t)^2$$

$$dY = \sigma d\tilde{W}(t) - \frac{1}{2} \sigma^2 t$$

$$d\ln(Z_t) = \sigma d\tilde{W}(t) - \frac{1}{2} \sigma^2 t$$

$$Z_t = Z_0 \exp \left\{ \sigma d\tilde{W}(t) - \frac{1}{2} \sigma^2 t \right\}$$

Heureka!

lets reunite

$$C_0 = S_1(0) \mathbb{E} \left[ \max \{ Z_T - 1, 0 \} \right]$$

$$\omega = S_1(v) \tilde{E} \left[ (z_T - 1) \mathbb{1}_{\{z_T \geq 1\}} \right]$$

$$\omega = S_1(v) \tilde{\Xi} \begin{bmatrix} z_1 & 1_{\{z_1 \geq 1\}} & -1_{\{z_1 \geq 1\}} \end{bmatrix}$$

$$C_0 = S_1(0) \tilde{E} \begin{bmatrix} z_T & \mathbb{1}_{\{z_T \geq 1\}} \end{bmatrix} - S_1(0) \tilde{E} \begin{bmatrix} \mathbb{1}_{\{z_T \geq 1\}} \end{bmatrix}$$

$$C_0 = S_1(C_0) \mathbb{E} \left[ z_T \mathbb{1}_{\{z_T \geq 1\}} \right] - S_1(C_0) \tilde{P}(z_T \geq 1)$$

Objective:  $\mathbb{E}[Z_T \mathbb{1}_{\{Z_0 > 1\}}]$  into  $\mathbb{E}[L_T \mathbb{1}_{\{Z_0 > 1\}}]$

$$\tilde{\mathbb{E}} \left[ Z_T \mathbb{1}_{\{Z_0 > 1\}} \right]$$

If we have some "vnikirni" derivative, this gives us a new measure.

$$\tilde{\mathbb{E}} \left[ L_T \mathbb{1}_{\{Z_0 > 1\}} \right] = \mathbb{E} \left[ \mathbb{1}_{\{Z_0 > 1\}} \right]$$

↑  
new measure.

transform the measure.

If we use the variables at  $t$ .

$$Z_t = Z_0 \exp \left\{ \sigma \tilde{W}(t) - \frac{1}{2} \sigma^2 t \right\}$$

Instead of  $Z_T$ , let's use  $\frac{Z_T}{Z_0}$

$$L_T = \frac{Z_T}{Z_0} = \exp \left\{ \sigma \tilde{W}(T) - \frac{1}{2} \sigma^2 T \right\}$$

now we can identify that the kernel is  $\sigma$

$$\varphi_t = \sigma$$

$$d\tilde{W}(t) = \sigma dt + d\bar{W}(t)$$

↑                    ↑                    |  
original under risk free measure  
                         |  
                         the kernel                    |

new brownian motion

we have new dynamics for  $Z_t$  under this measure



Before, in the risk neutral measure:

$$Z_T = Z_0 \exp \left\{ \sigma d\tilde{W}(T) - \frac{1}{2} \sigma^2 T \right\}$$

After, in the new measure:

$$Z_T = Z_0 \exp \left\{ \sigma d\tilde{W}(T) + \frac{1}{2} \sigma^2 T \right\}$$

↑  
we still have this...

the goal is to  $\mathbb{E} \left[ Z_T \mathbb{1}_{\{Z_T \geq 1\}} \right]$  into  $\mathbb{E} \left[ L_T \mathbb{1}_{\{Z_T \geq 1\}} \right]$

$$\begin{aligned} \tilde{\mathbb{E}} \left[ Z_T \mathbb{1}_{\{Z_T \geq 1\}} \right] &= Z_0 \tilde{\mathbb{E}} \left[ \frac{Z_T}{Z_0} \mathbb{1}_{\{Z_T \geq 1\}} \right] \\ &= Z_0 \tilde{\mathbb{E}} \left[ L_T \mathbb{1}_{\{Z_T \geq 1\}} \right] \\ &= Z_0 \tilde{\mathbb{E}} \left[ \mathbb{1}_{\{Z_T \geq 1\}} \right] \end{aligned}$$

$$\begin{aligned} \omega &= S_1(0) \tilde{\mathbb{E}} \left[ Z_T \mathbb{1}_{\{Z_T \geq 1\}} \right] - S_1(0) \tilde{\mathbb{E}} \left[ \mathbb{1}_{\{Z_T \geq 1\}} \right] \\ &= S_1(0) Z_0 \tilde{\mathbb{E}} \left[ \mathbb{1}_{\{Z_T \geq 1\}} \right] - S_1(0) \tilde{\mathbb{E}} \left[ \mathbb{1}_{\{Z_T \geq 1\}} \right] \end{aligned}$$

$$= S_1(0) Z_0 \underset{\substack{\uparrow \\ \text{new measure}}}{\bar{\mathbb{P}}} (Z_T \geq 1) - S_1(0) \underset{\substack{\uparrow \\ \text{risk neutral measure}}}{\tilde{\mathbb{P}}} (Z_T \geq 1)$$

$$\bar{P}(Z_T \geq 1)$$

(under the measure)

$$Z_t \geq 1$$

$$Z_0 \exp\left\{\sigma d\bar{W}(t) + \frac{1}{2}\sigma^2 t\right\} \geq 1$$

$$\sigma d\bar{W}(t) \geq \ln\left(\frac{1}{Z_0}\right) - \frac{1}{2}\sigma^2 t$$

$$-\sigma d\bar{W}(t) \leq \ln(Z_0) + \frac{1}{2}\sigma^2 t$$

$$-d\bar{W}(t) \leq \frac{\ln(Z_0) + \frac{1}{2}\sigma^2 t}{\sigma}$$

What's the distribution of this?

$$\bar{P}(Z_T \geq 1) = \bar{P}\left(-d\bar{W}(t) \leq \frac{\ln(Z_0) + \frac{1}{2}\sigma^2 t}{\sigma}\right), \text{ remember that } -d\bar{W}(t) \sim \mathcal{N}(0, t)$$

$$= \bar{P}\left(\frac{-d\bar{W}(t)}{\sqrt{t}} \leq \frac{\ln(Z_0) + \frac{1}{2}\sigma^2 t}{\sqrt{t}\sigma}\right), \text{ let } d_1 := \frac{\ln(Z_0) + \frac{1}{2}\sigma^2 t}{\sqrt{t}\sigma}$$

$$= \Phi(d_1)$$

then:

where  $\Phi$  is the c.f.d of the normal distribution

$$\tilde{P}(Z_T \geq 1)$$

(under the risk free measure)

$$Z_t \geq 1$$

$$Z_0 \exp\left\{\sigma d\tilde{W}(t) - \frac{1}{2}\sigma^2 T\right\} \geq 1$$

$$\sigma d\tilde{W}(t) \geq \ln\left(\frac{1}{Z_0}\right) + \frac{1}{2}\sigma^2 T$$

$$-\sigma d\tilde{W}(t) \leq \ln(Z_0) - \frac{1}{2}\sigma^2 T$$

$$-d\tilde{W}(t) \leq \frac{\ln(Z_0) - \frac{1}{2}\sigma^2 T}{\sigma}$$

$$\tilde{P}(Z_T \geq 1) = \tilde{P}\left(-d\tilde{W}(t) \leq \frac{\ln(Z_0) - \frac{1}{2}\sigma^2 T}{\sigma}\right), \text{ remember that } -d\tilde{W}(t) \sim \mathcal{N}(0, t)$$

$$= \tilde{P}\left(\frac{-d\tilde{W}(t)}{\sqrt{T}} \leq \frac{\ln(Z_0) - \frac{1}{2}\sigma^2 T}{\sqrt{T}\sigma}\right), \text{ let } dz := \frac{\ln(Z_0) - \frac{1}{2}\sigma^2 T}{\sqrt{T}\sigma}$$

$$= \Phi(dz)$$

then:

where  $\Phi$  is the c.f.d of the normal distribution

$$\omega = S_1(0) Z_0 \bar{P}(Z_T \geq 1) - S_1(0) \tilde{P}(Z_T \geq 1)$$

$$= S_1(0) z_0 \Phi(d_1) - S_1(0) \Phi(d_2)$$

$$= S_1(0) \frac{S_2(0)}{S_1(0)} \Phi(d_1) - S_1(0) \Phi(d_2)$$

$$= S_2(0) \Phi(d_1) - S_1(0) \Phi(d_2)$$

$$C_0 = S_2(0) \Phi(d_1) - S_1(0) \Phi(d_2),$$

where:

$$d_1 = \frac{\ln(z_0) + \frac{1}{2} \sigma^2 T}{\sqrt{T} \sigma} \quad \text{and} \quad d_2 = \frac{\ln(z_0) - \frac{1}{2} \sigma^2 T}{\sqrt{T} \sigma}$$

Ne amse

wo wowowo

pls like and  
subscribe if you  
liked this